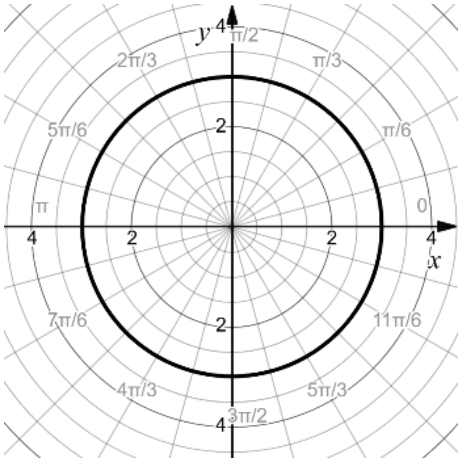
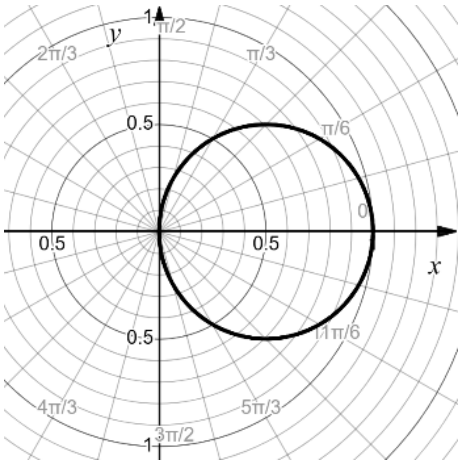
1) a)



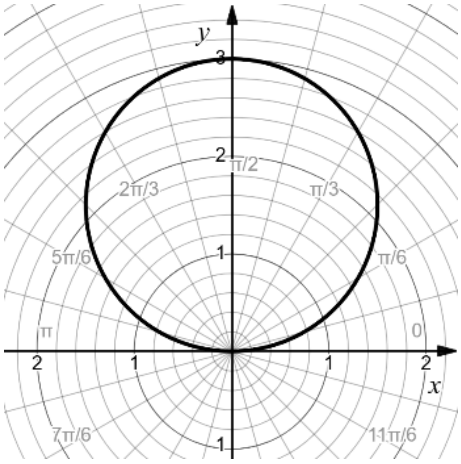
1) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	$\sqrt{2}$	0	$-\sqrt{2}$	-1	$-\sqrt{2}$	0	$\sqrt{2}$	1



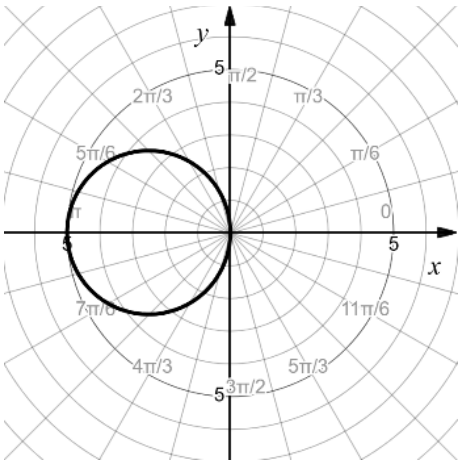
1) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	$\frac{3\sqrt{2}}{2}$	3	$\frac{3\sqrt{2}}{2}$	0	$-\frac{3\sqrt{2}}{2}$	-3	$-\frac{3\sqrt{2}}{2}$	0



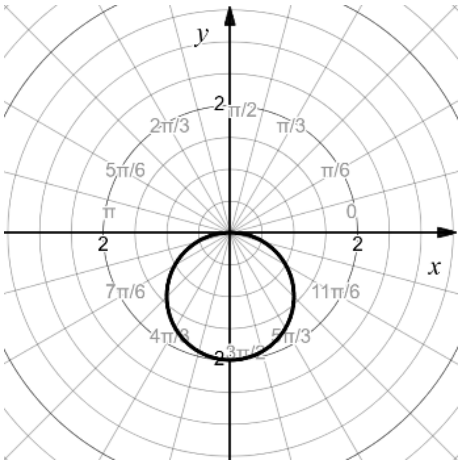
1) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-5	$-\frac{5\sqrt{2}}{2}$	0	$\frac{5\sqrt{2}}{2}$	5	$\frac{5\sqrt{2}}{2}$	0	$-\frac{5\sqrt{2}}{2}$	-5



1) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	$-\sqrt{2}$	-2	$-\sqrt{2}$	0	$\sqrt{2}$	2	$\sqrt{2}$	0



2) a)

$r = 5$

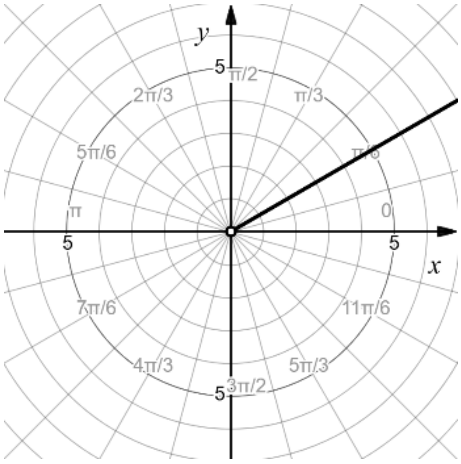
2) b)

$r = 4\cos\theta$

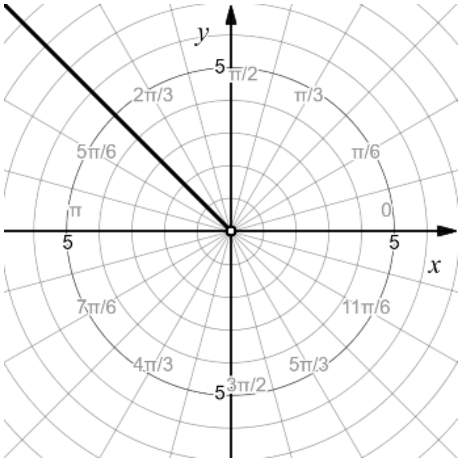
2) c)

$r = 3\sin\theta$

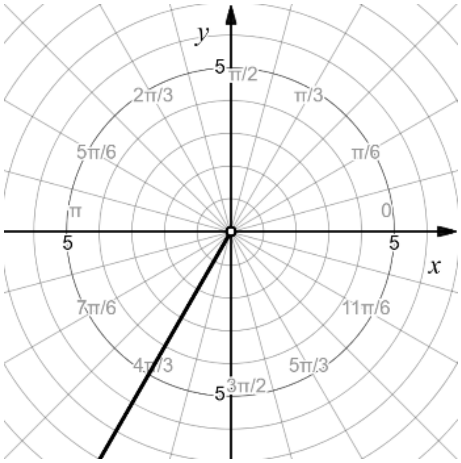
3) a)



3) b)



3) c)



4) a)

$\theta = \frac{\pi}{3}$

4) b)

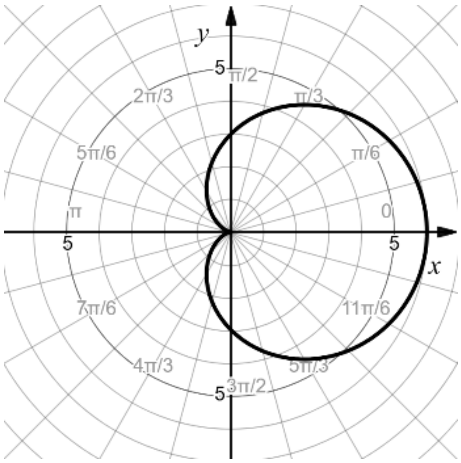
$\theta = \frac{5\pi}{6}$

4) c)

$\theta = -\frac{4\pi}{3}$

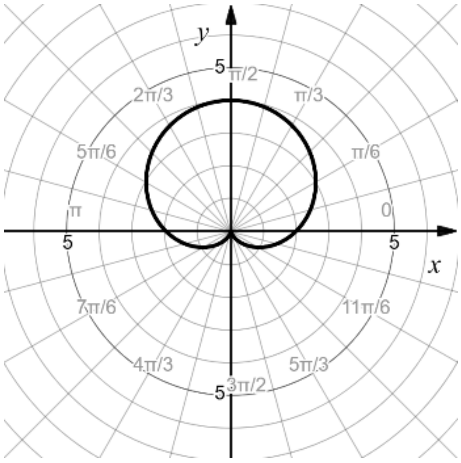
5) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	6	5.12	3	0.879	0	0.879	3	5.12	6



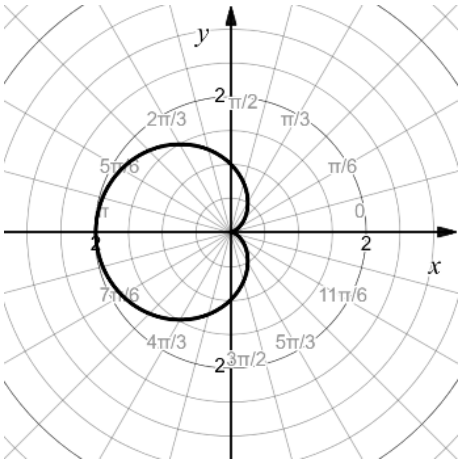
5) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	2	3.41	4	3.41	2	0.586	0	0.586	2



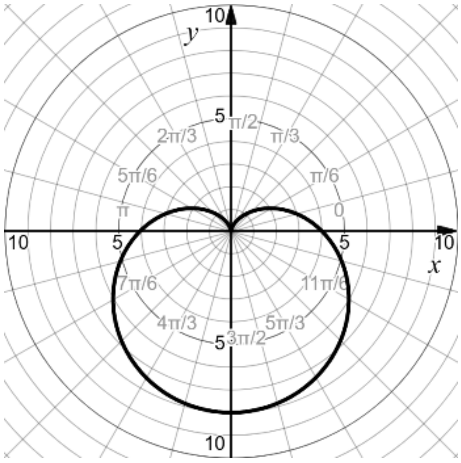
5) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0.293	1	1.71	2	1.71	1	0.293	0



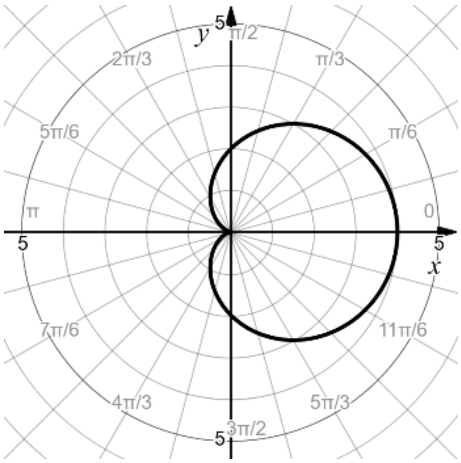
5) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	1.17	0	1.17	4	6.83	8	6.83	4



5) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	3.41	2	0.586	0	0.586	2	3.41	4



6) a)

$r = 2.5(1 + \sin\theta)$

6) b)

$r = 1.5(1 - \cos\theta)$

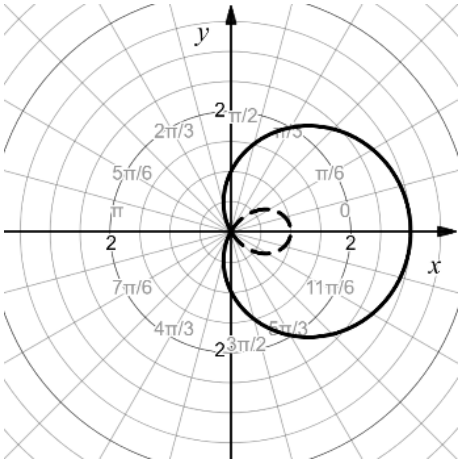
6) c)

$r = 2(1 + \cos\theta)$

7) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	2.41	1	-0.414	-1	-0.414	1	2.41	3

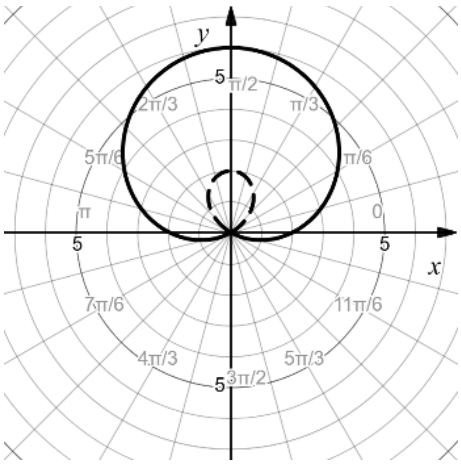
$0 = 1 + 2\cos\theta$ has solutions $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$ so these are the two values between which r is negative.



7) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	2	4.83	6	4.83	2	-0.828	-2	-0.828	2

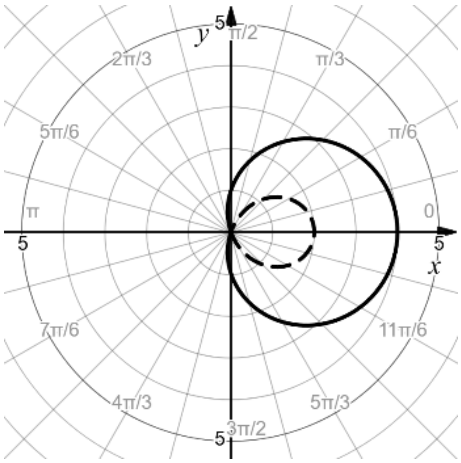
$0 = 2 + 4\sin\theta$ has solutions $\theta = \frac{7\pi}{6}$ and $\theta = \frac{11\pi}{6}$ so these are the two values between which r is negative.



7) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	3.12	1	-1.12	-2	-1.12	1	3.12	4

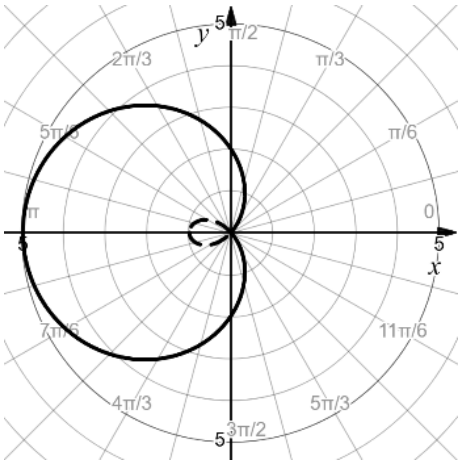
$0 = 1 + 3\cos\theta$ has solutions $\theta = 1.91$ and $\theta = 4.37$ so these are the two values between which r is negative.



7) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-1	-0.121	2	4.12	5	4.12	2	-0.12	-1

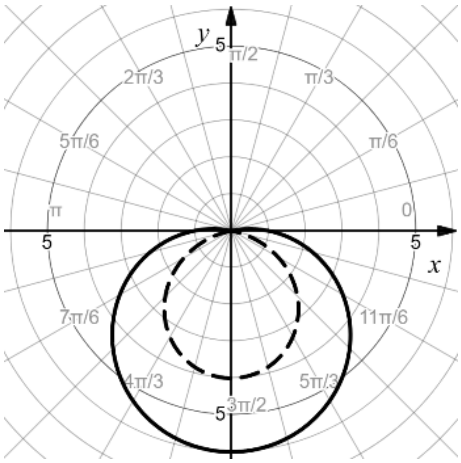
$0 = 2 - 3\cos\theta$ has solutions $\theta = 0.841$ and $\theta = 5.44$ so r is negative up to 0.841 and after 5.44



7) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-2.54	-4	-2.54	1	4.54	6	4.54	1

$0 = 1 - 5\sin\theta$ has solutions $\theta = 0.201$ and $\theta = 2.94$ so these are the two values between which r is negative.



8) a)

Maximum $r = 5$

Internal loop has $r = -1$

So sine with amplitude 3

$r = 2 - 3\sin\theta$

8) b)

Maximum $r = 4$

Internal loop has $r = -2$

So cos with amplitude 3

$r = 1 + 3\cos\theta$

8) c)

Maximum $r = 3$

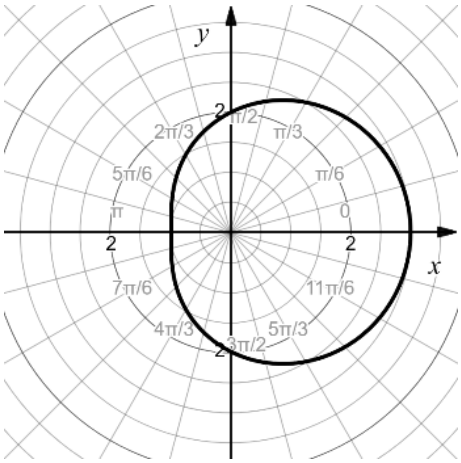
Internal loop has $r = -1$

So cos with amplitude 2

$r = 1 - 2\cos\theta$

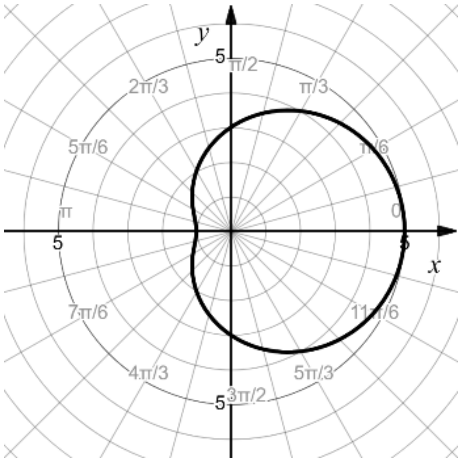
9) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	2.71	2	1.29	1	1.29	2	2.71	3



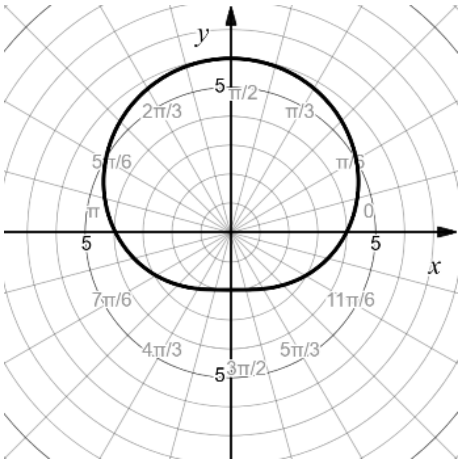
9) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	5	4.41	3	1.59	1	1.59	3	4.41	5



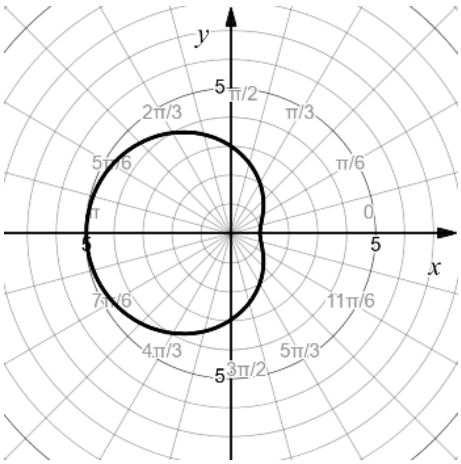
9) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	5.41	6	5.41	4	2.59	2	2.59	4



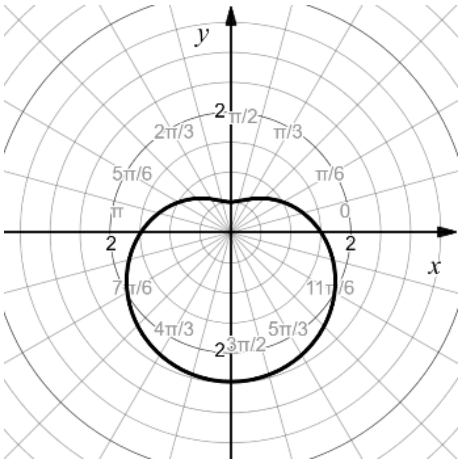
9) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	1.59	3	4.41	5	4.41	3	1.59	1



9) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1.5	0.793	0.5	0.793	1.5	2.21	2.5	2.21	1.5



10) a)

Maximum $r = 3$

Minimum $r = 1$

So sin with amplitude 1

$r = 2 + \sin\theta$

10) b)

Maximum $r = 4$

Minimum $r = 2$

So cos with amplitude 1

$r = 3 + \cos\theta$

10) c)

Maximum $r = 6$

Minimum $r = 1$

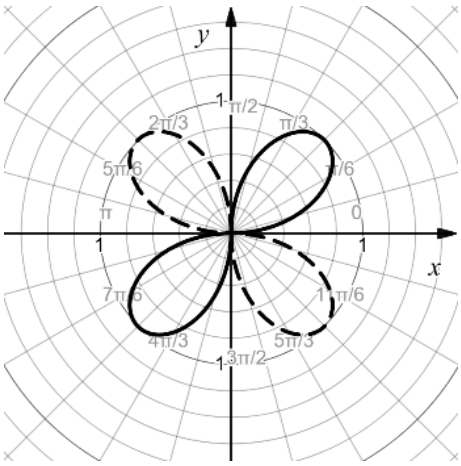
So cos with amplitude 2.5

$r = 3.5 - 2.5\cos\theta$

11) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	1	0	-1	0	1	0	-1	0

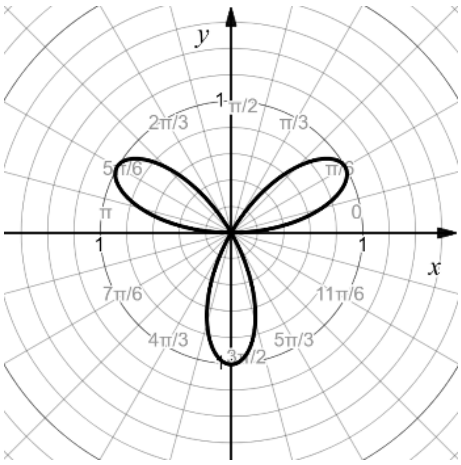
$0 = \sin 2\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{2}$, $\theta = \pi$, $\theta = \frac{3\pi}{2}$ and $\theta = 2\pi$ so these are the five values at which r swaps between positivity.



11) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0.707	-1	0.707	0	-0.707	1	-0.707	0

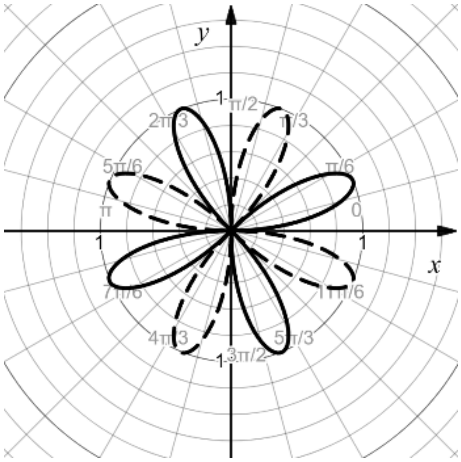
$0 = \sin 3\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{3}$, $\theta = \frac{2\pi}{3}$, $\theta = \pi$, $\theta = \frac{4\pi}{3}$, $\theta = \frac{5\pi}{3}$ and $\theta = 2\pi$ so these are the seven values at which r swaps between positivity.
However, the 6 'petals' overlap leaving 3 'petals'.



11) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0	0	0	0	0	0	0	0

$0 = \sin 4\theta$ has solutions given in the table above, these are the nine values at which r swaps between positivity.

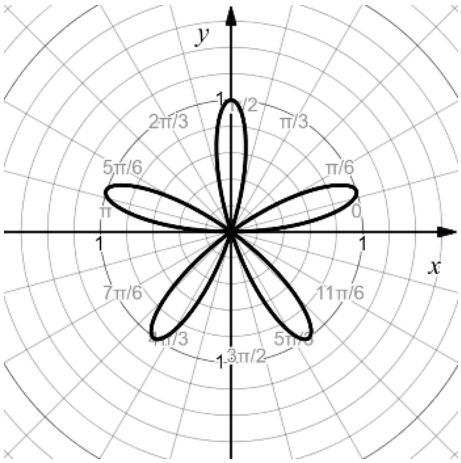


11) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	-0.707	1	-0.707	0	0.707	-1	0.707	0

$0 = \sin 5\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{5}$, $\theta = \frac{2\pi}{5}$, …, $\theta = \frac{7\pi}{5}$, $\theta = 2\pi$ so these are the eleven values at which r swaps between positivity.

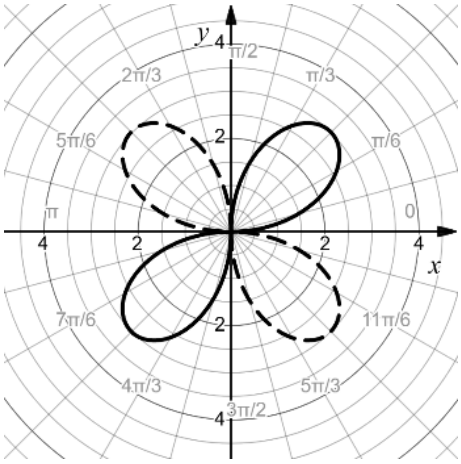
However, the 10 'petals' overlap leaving 5 'petals'.



11) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	3	0	-3	0	3	0	-3	0

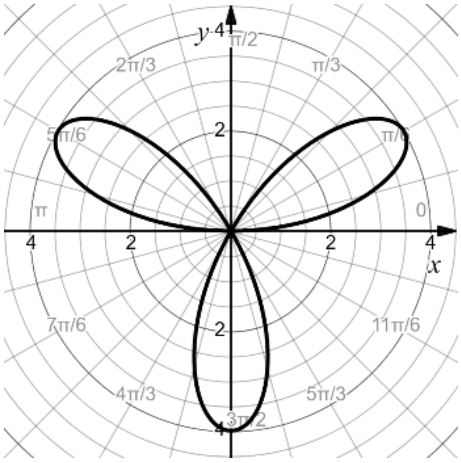
$0 = 3\sin 2\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{2}$, $\theta = \pi$, $\theta = \frac{3\pi}{2}$ and $\theta = 2\pi$ so these are the five values at which r swaps between positivity.



11) f)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2.83	-4	2.83	0	-2.83	4	-2.83	0

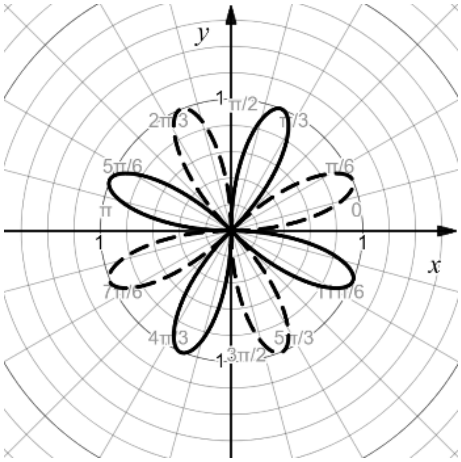
$0 = 4\sin 3\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{3}$, $\theta = \frac{2\pi}{3}$, $\theta = \pi$, $\theta = \frac{4\pi}{3}$, $\theta = \frac{5\pi}{3}$ and $\theta = 2\pi$ so these are the seven values at which r swaps between positivity.
However, the 6 'petals' overlap leaving 3 'petals'.



11) g)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0	0	0	0	0	0	0	0

$0 = -\sin 4\theta$ has solutions given in the table above, these are the nine values at which r swaps between positivity, starting with negative values of r

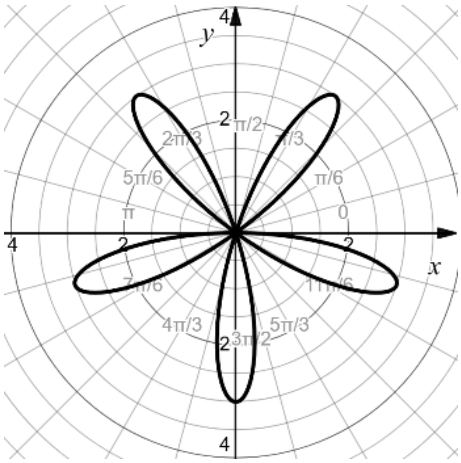


11) h)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2.12	-3	2.12	0	-2.12	3	-2.12	0

$0 = -3\sin 5\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{5}$, $\theta = \frac{2\pi}{5}$, …, $\theta = \frac{7\pi}{5}$, $\theta = 2\pi$ so these are the eleven values at which r swaps between positivity.

However, the 10 'petals' overlap leaving 5 'petals'.



12) a)

k

12) b)

$-k$

12) c)

a times

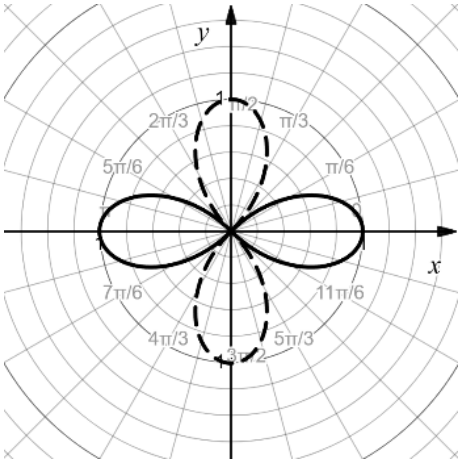
12) d)

$2a$ times

13) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	0	-1	0	1	0	-1	0	1

$0 = \cos 2\theta$ has solutions $\theta = \frac{\pi}{4}$, $\theta = \frac{3\pi}{4}$, $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$ so these are the four values at which r swaps between positivity.

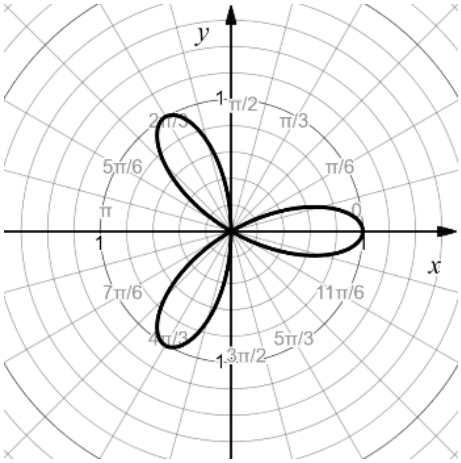


13) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-0.707	0	0.707	-1	0.707	0	-0.707	1

$0 = \cos 3\theta$ has solutions $\theta = \frac{\pi}{6}$, $\theta = \frac{3\pi}{6}$, $\theta = \frac{5\pi}{6}$, $\theta = \frac{7\pi}{6}$, $\theta = \frac{9\pi}{6}$ and $\theta = \frac{11\pi}{6}$ so these are the six values at which r swaps between positivity.

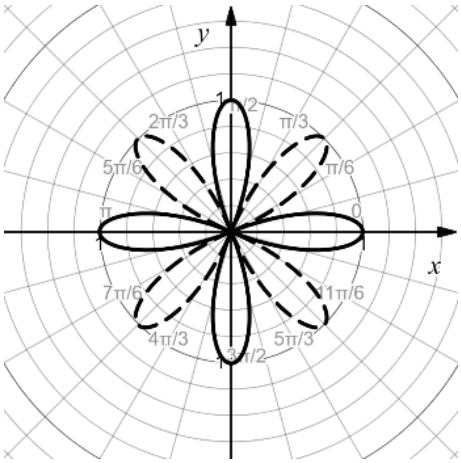
The 3 'petals' overlap so that there are both positive and negative values of r on this graph.



13) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-1	1	-1	1	-1	1	-1	1

$0 = \cos 4\theta$ has solutions $\theta = \frac{\pi}{8}$, $\theta = \frac{3\pi}{8}$, $\theta = \frac{5\pi}{8}$, …, $\theta = \frac{13\pi}{8}$, $\theta = \frac{15\pi}{8}$ so these are the eight values at which r swaps between positivity.

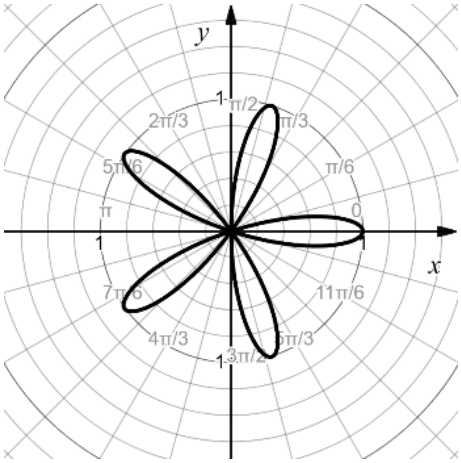


13) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-0.707	0	0.707	-1	0.707	0	-0.707	1

$0 = \cos 5\theta$ has solutions $\theta = \frac{\pi}{10}$, $\theta = \frac{3\pi}{10}$, $\theta = \frac{5\pi}{10}$, …, $\theta = \frac{17\pi}{10}$, $\theta = \frac{19\pi}{10}$ so these are the ten values at which r swaps between positivity.

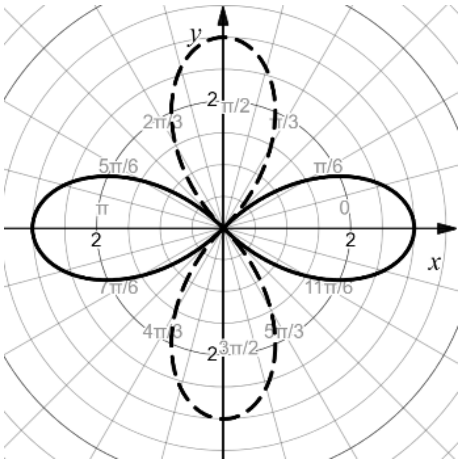
The 5 'petals' overlap so that there are both positive and negative values of r on this graph.



13) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	0	-3	0	3	0	-3	0	3

$0 = 3\cos 2\theta$ has solutions $\theta = \frac{\pi}{4}$, $\theta = \frac{3\pi}{4}$, $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$ so these are the four values at which r swaps between positivity.

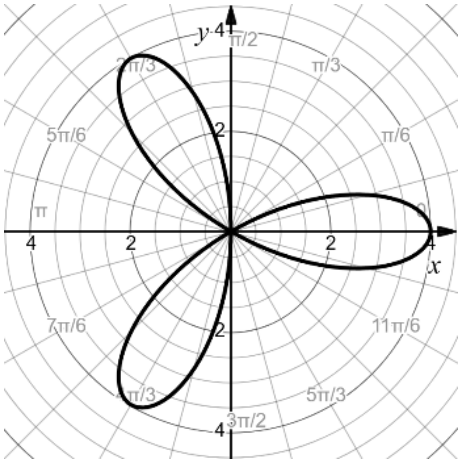


13) f)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	-2.83	0	2.83	-4	2.83	0	-2.83	4

$0 = 4\cos 3\theta$ has solutions $\theta = \frac{\pi}{6}$, $\theta = \frac{3\pi}{6}$, $\theta = \frac{5\pi}{6}$, $\theta = \frac{7\pi}{6}$, $\theta = \frac{9\pi}{6}$ and $\theta = \frac{11\pi}{6}$ so these are the six values at which r swaps between positivity.

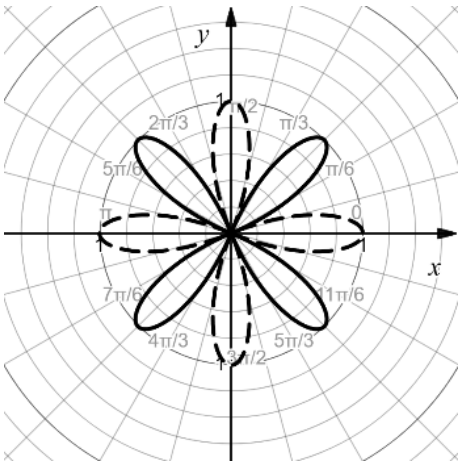
The 3 'petals' overlap so that there are both positive and negative values of r on this graph.



13) g)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-1	1	-1	1	-1	1	-1	1	-1

$0 = \cos 4\theta$ has solutions $\theta = \frac{\pi}{8}$, $\theta = \frac{3\pi}{8}$, $\theta = \frac{5\pi}{8}$, …, $\theta = \frac{13\pi}{8}$, $\theta = \frac{15\pi}{8}$ so these are the eight values at which r swaps between positivity.

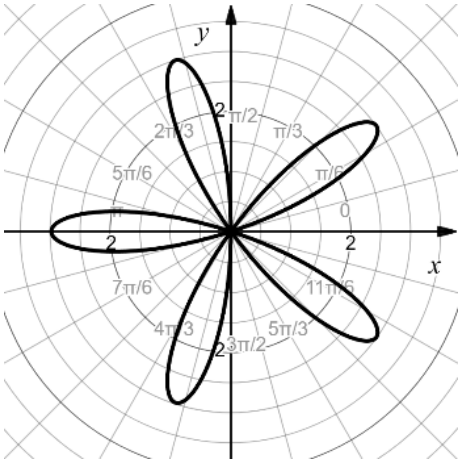


13) h)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-3	2.12	0	-2.12	3	-2.12	0	2.12	-3

$0 = -3\cos 5\theta$ has solutions $\theta = \frac{\pi}{10}$, $\theta = \frac{3\pi}{10}$, $\theta = \frac{5\pi}{10}$, …, $\theta = \frac{17\pi}{10}$, $\theta = \frac{19\pi}{10}$ so these are the ten values at which r swaps between positivity.

The 5 'petals' overlap so that there are both positive and negative values of r on this graph.



14) a)

k

14) b)

$-k$

14) c)

a times

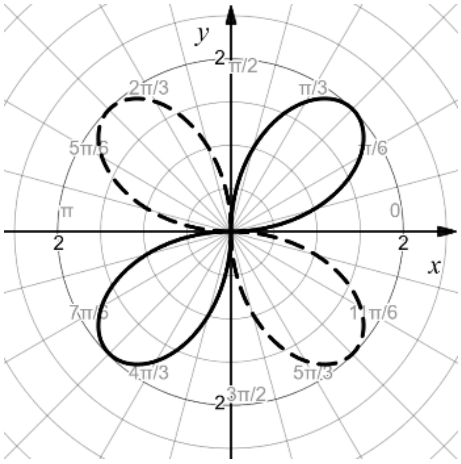
14) d)

$2a$ times

15) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2	0	-2	0	2	0	-2	0

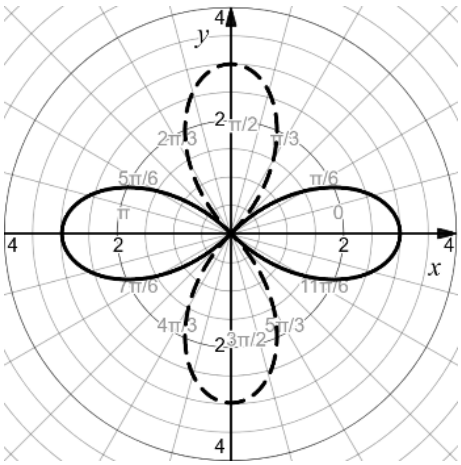
$0 = 2\sin 2\theta$ has solutions $\theta = \frac{\pi}{2}$, $\theta = \pi$ and $\theta = \frac{3\pi}{2}$ so these are the three values at which r swaps between positivity.



15) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	0	-3	0	3	0	-3	0	3

$0 = 3\cos 2\theta$ has solutions $\theta = \frac{\pi}{4}$, $\theta = \frac{3\pi}{4}$, $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$ so these are the four values at which r swaps between positivity.

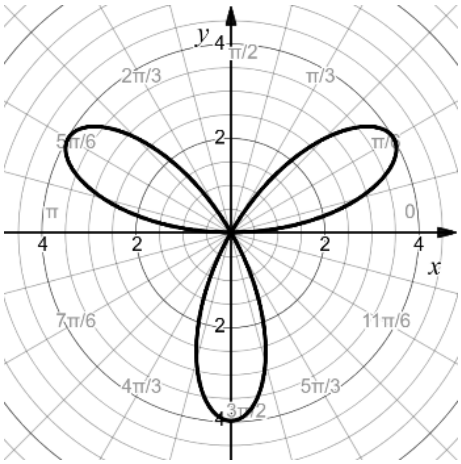


15) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2.82	-4	2.82	0	-2.82	4	-2.82	0

$0 = 4\sin 3\theta$ has solutions $\theta = \frac{\pi}{3}, \theta = \frac{2\pi}{3}, \theta = \pi, \theta = \frac{4\pi}{3}$ and $\theta = \frac{5\pi}{3}$ so these are the five values at which r swaps between positivity.

However, the 6 'petals' overlap leaving 3 'petals'.

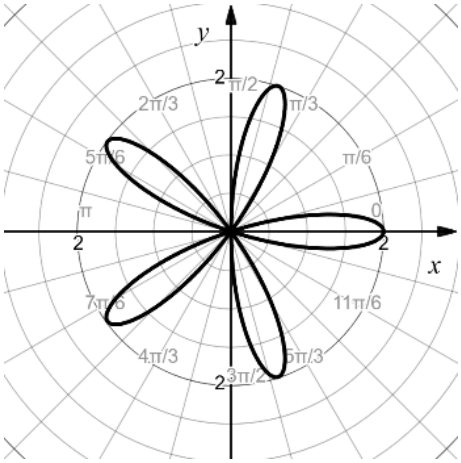


15) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	2	-1.41	0	1.41	-2	1.41	0	-1.41	2

$0 = 2\cos 5\theta$ has solutions $\theta = \frac{\pi}{10}, \theta = \frac{3\pi}{10}, \theta = \frac{5\pi}{10}, \theta = \frac{7\pi}{10}, \theta = \frac{9\pi}{10}, \theta = \frac{11\pi}{10}, \theta = \frac{13\pi}{10}, \theta = \frac{15\pi}{10}, \theta = \frac{17\pi}{10}$ and $\theta = \frac{19\pi}{10}$ so these are the ten values at which r swaps between positivity.

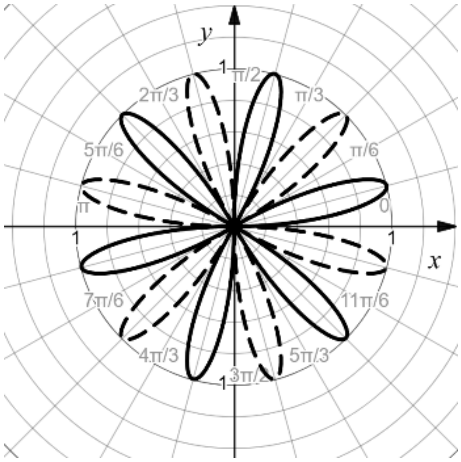
However, the 10 'petals' overlap leaving 5 'petals'.



15) e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	-1	0	1	0	-1	0	1	0

$0 = \sin 6\theta$ has solutions $\theta = \frac{\pi}{6}, \theta = \frac{2\pi}{6}, \theta = \frac{3\pi}{6}, \theta = \frac{4\pi}{6}, \theta = \frac{5\pi}{6}, \theta = \pi, \theta = \frac{7\pi}{6}, \theta = \frac{8\pi}{6}, \theta = \frac{9\pi}{6}, \theta = \frac{10\pi}{6}, \theta = \frac{11\pi}{6}$ and $\theta = 2\pi$ so these are the twelve values at which r swaps between positivity.



16) a)

Cosine with 5 petals:

$r = 4\cos 5\theta$

16) b)

Sine with 16 petals:

$r = 2\sin 8\theta$

16) c)

Sine with 4 petals:

$r = 4\sin 2\theta$

16) d)

Cosine with 8 petals:

$r = 5\cos 4\theta$

16) e)

Cosine with 4 petals:

$r = 3\cos 2\theta$

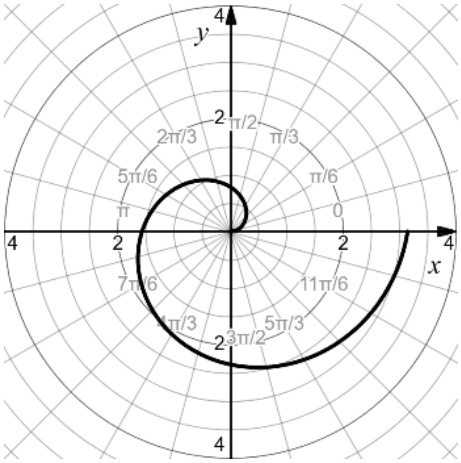
16) f)

Sine with 3 petals:

$r = 5\sin 3\theta$

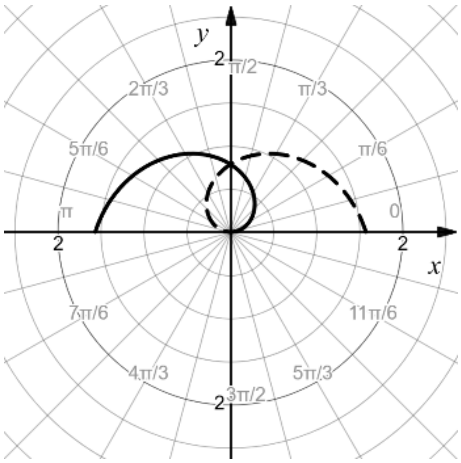
17) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0.393	0.785	1.19	1.57	1.96	2.36	2.75	3.14



17) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-1.57	-1.18	-0.785	-0.392	0	0.392	0.785	1.18	1.57



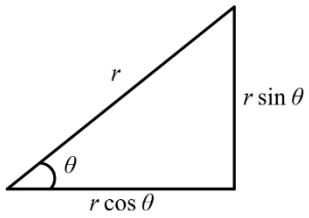
18)

The gaps between the spirals on the curve become closer together.

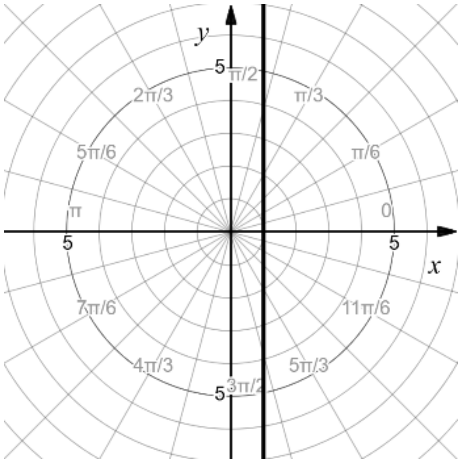
19) a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	$\sqrt{2}$		$-\sqrt{2}$	-1	$-\sqrt{2}$		$\sqrt{2}$	1

By considering the right-angle triangle:



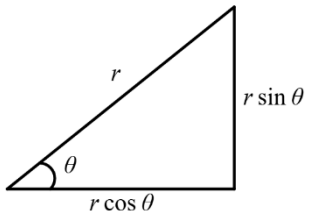
Since $r\cos\theta = 1$ the horizontal distance is always 1



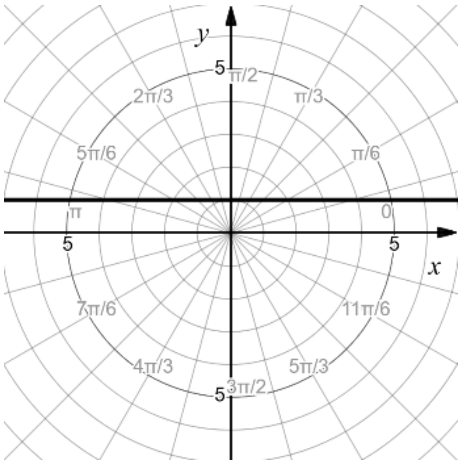
19) b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$		$\sqrt{2}$	1	$\sqrt{2}$		$-\sqrt{2}$	-1	$-\sqrt{2}$	

By considering the right-angle triangle:



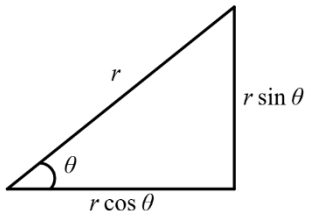
Since $r\sin\theta = 1$ the vertical distance is always 1



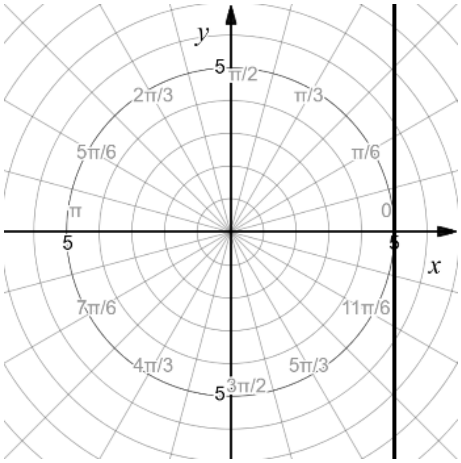
19) c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	5	$5\sqrt{2}$		$-5\sqrt{2}$	-5	$-5\sqrt{2}$		$5\sqrt{2}$	5

By considering the right-angle triangle:



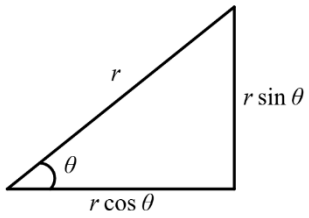
Since $r\cos\theta = 5$ the horizontal distance is always 5



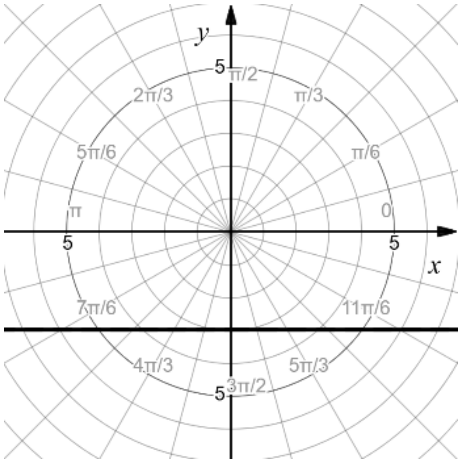
19) d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$		$-3\sqrt{2}$	-3	$-3\sqrt{2}$		$3\sqrt{2}$	3	$3\sqrt{2}$	

By considering the right-angle triangle:



Since $r\sin\theta = -3$ the vertical distance is always -3



20) a)

$r = \frac{3}{\cos\theta}$

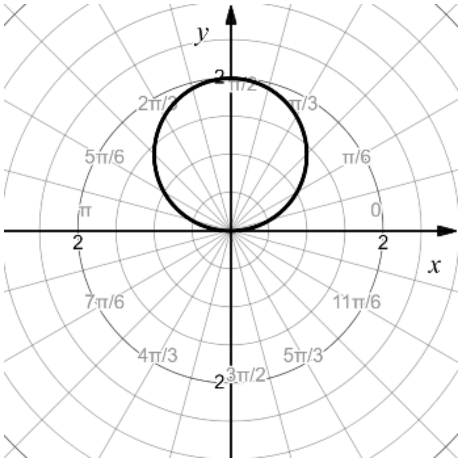
20) b)

$r = \frac{4}{\sin\theta}$

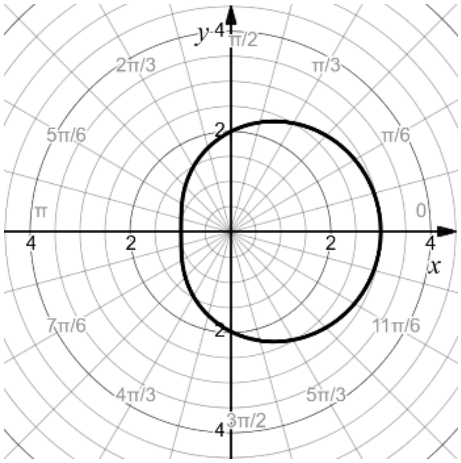
20) c)

$r = -\frac{2}{\cos\theta}$

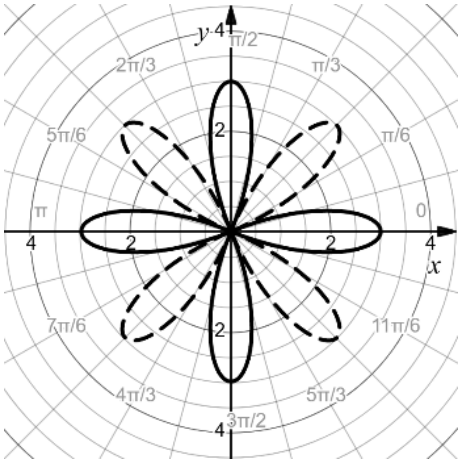
21) a)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
2
2
2
2

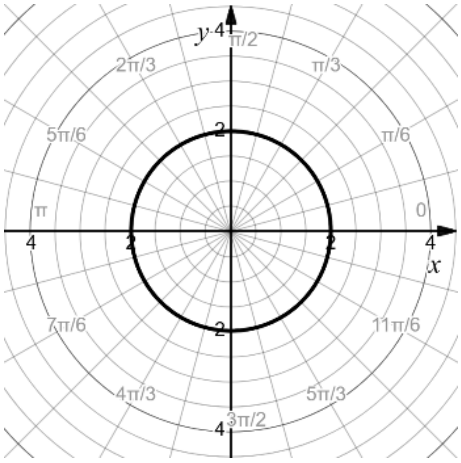
21) b)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
4
2
2
4
4
2
2
4

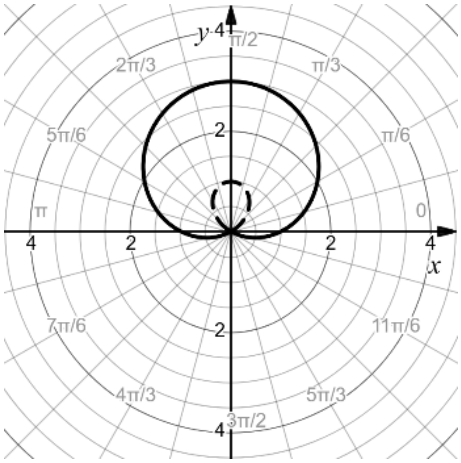
21) c)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
4
2
2
4
4
2
2
4

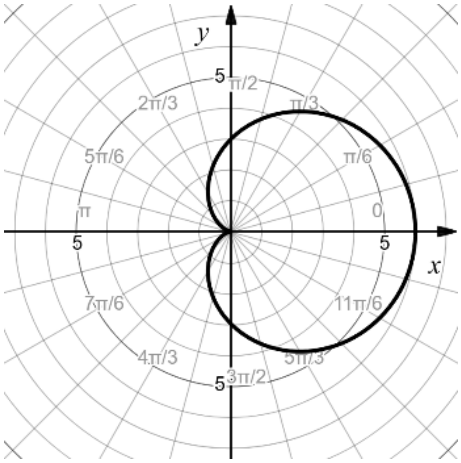
21) d)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
2
4

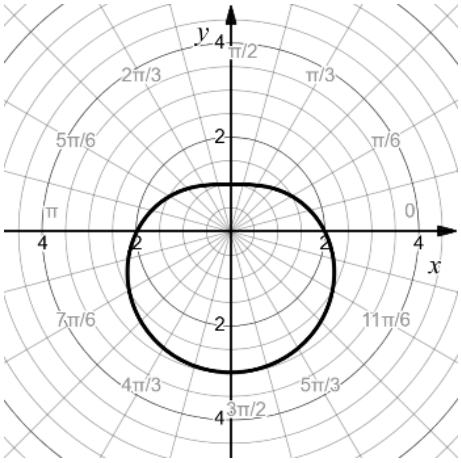
21) e)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
2
4

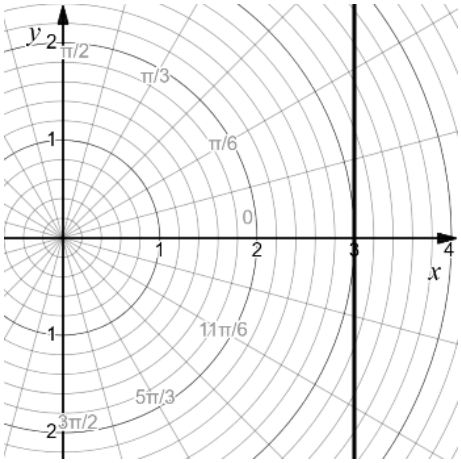
21) f)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
5

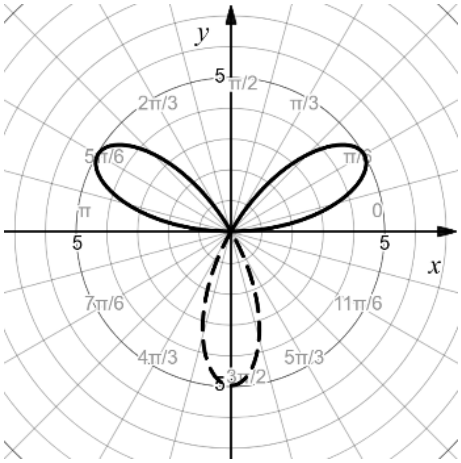
21) g)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
2
4

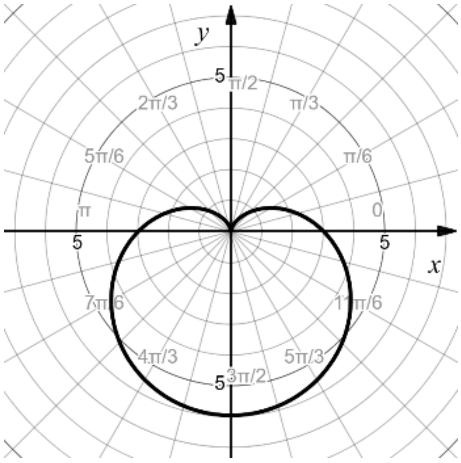
21) h)


y
x
π/2
π/3
π/6
0
11π/6
5π/3
3π/2
1
2
3
4

21) i)


y
x
π/2
π/3
2π/3
π
0
7π/6
11π/6
4π/3
5π/3
5

21) j)



22) a)

$r = \frac{a}{\sin\theta}$ where $\theta \neq n\pi$ for $n \in \mathbb{Z}$

22) b)

$r = \frac{2a}{\cos\theta}$ where $\theta \neq \frac{n\pi}{2}$ for $n \in \mathbb{Z}$

22) c)

$r = a\cos\theta$

22) d)

$r = 2a\sin\theta$

22) e)

$r = a + a\cos\theta$ or $r = a - a\cos\theta$

Both of these pass through $(0, a)$

$\therefore r = a + a\cos\theta$ or $r = a - a\cos\theta$

22) f)

$r = \frac{a}{2} + \frac{a}{2}\sin\theta$ or $r = \frac{a}{2} - \frac{a}{2}\sin\theta$

Both of these pass through $\left(\frac{a}{2}, 0\right)$

$\therefore r = \frac{a}{2} + \frac{a}{2}\sin\theta$ or $r = \frac{a}{2} - \frac{a}{2}\sin\theta$

23) a)

Sine with 8 petals:

$r = 4\sin 4\theta$

23) b)

$r = 2.5(1 - \sin\theta)$

23) c)

Maximum $r = 5$

Internal loop has $r = -1$

So cosine with amplitude 3

$r = 2 + 3\cos\theta$ or $r = -2 + 3\cos\theta$

23) d)

Cosine with 7 petals:

$r = 5\cos 7\theta$

24)

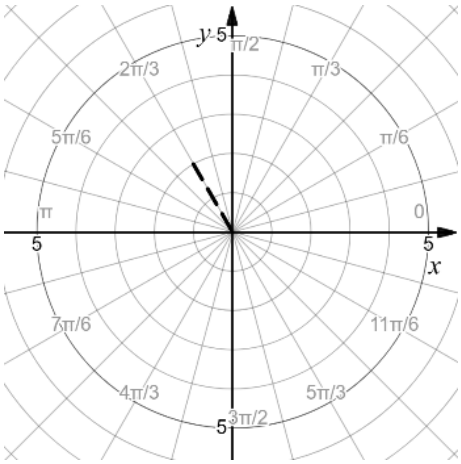
$r = \frac{2}{\sin\left(\theta - \frac{\pi}{6}\right)}$

$\sin\left(\theta - \frac{\pi}{6}\right)$ for $\frac{\pi}{6}$ to $\frac{4\pi}{6}$ increases from 0 to 1

$\sin\left(\theta - \frac{\pi}{6}\right)$ for $\frac{4\pi}{6}$ to $\frac{7\pi}{6}$ decreases from 1 to 0

At $\theta = \frac{2\pi}{3}$ then $r = 2$

So the shortest distance to the line is $r = 2$ which is at angle $\theta = \frac{2\pi}{3}$

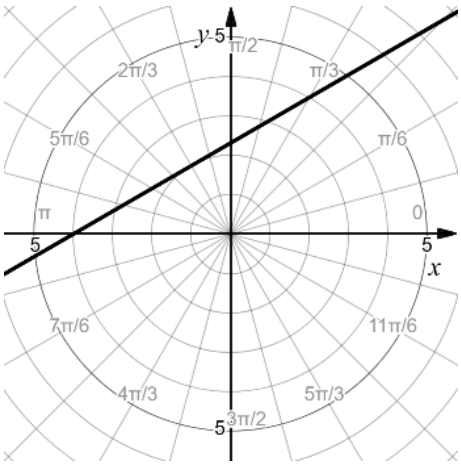


A straight line where:

$\boldsymbol{\theta}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	$\frac{6\pi}{6}$
$\boldsymbol{r}$	4	$\frac{4\sqrt{3}}{3}$	2	$\frac{4\sqrt{3}}{3}$	4

At $\theta = \pi$ then $r = 4$

At $\theta = \frac{\pi}{2}$ then $r = \frac{4\sqrt{3}}{3}$



25)

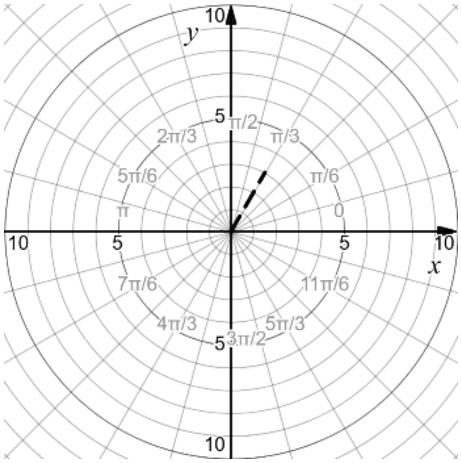
$r = \frac{3}{\cos\left(\theta - \frac{\pi}{3}\right)}$

$\cos\left(\theta - \frac{\pi}{3}\right)$ for $-\frac{\pi}{6}$ to $\frac{\pi}{3}$ decreases from 1 to 0

$\cos\left(\theta - \frac{\pi}{3}\right)$ for $\frac{\pi}{3}$ to $\frac{5\pi}{6}$ increases from 0 to 1

At $\theta = \frac{\pi}{3}$ then $r = 3$

So the shortest distance to the line is $r = 3$ which is at angle $\theta = \frac{\pi}{3}$

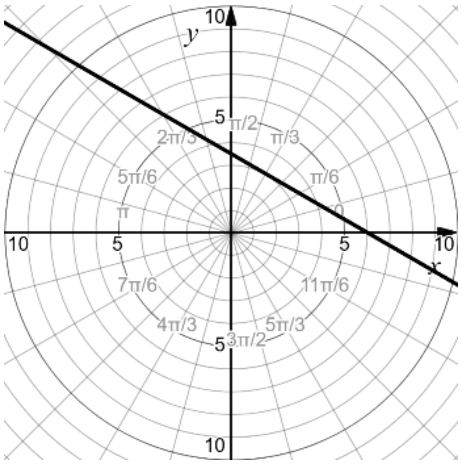


A straight line where:

$\boldsymbol{\theta}$	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$
$\boldsymbol{r}$	6	$2\sqrt{3}$	3	$2\sqrt{3}$	6

At $\theta = 0$ then $r = 6$

At $\theta = \frac{\pi}{2}$ then $r = 2\sqrt{3}$



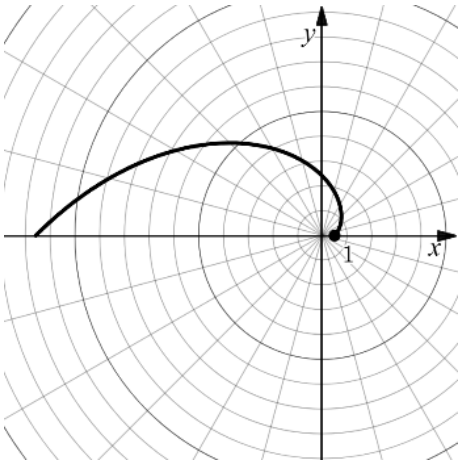
26) a)

$r = \sinh\theta + \cosh\theta = \frac{1}{2}\left(e^{\theta} - e^{-\theta}\right) + \frac{1}{2}\left(e^{\theta} + e^{-\theta}\right) = e^{\theta}$

When $\theta = 0$, $r = 1$

When $\theta = \frac{\pi}{2}$, $r = e^{\frac{\pi}{2}}$

When $\theta = \pi$, $r = e^{\pi}$



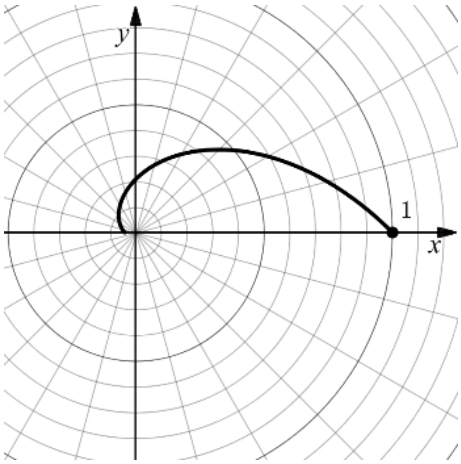
26) b)

$r = \cosh\theta - \sinh\theta = \frac{1}{2}\left(e^{\theta} + e^{-\theta}\right) - \frac{1}{2}\left(e^{\theta} - e^{-\theta}\right) = e^{-\theta}$

When $\theta = 0$, $r = 1$

When $\theta = \frac{\pi}{2}$, $r = e^{-\frac{\pi}{2}}$

When $\theta = \pi$, $r = e^{-\pi}$



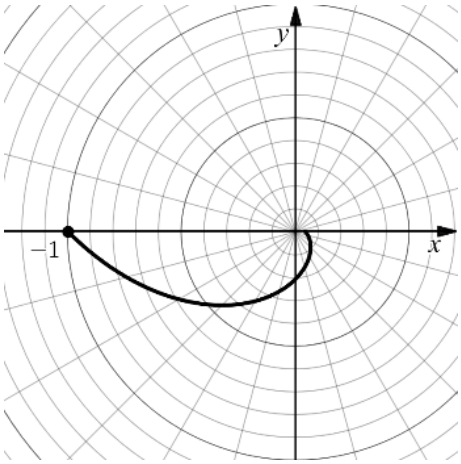
26) c)

$r = \sinh\theta - \cosh\theta = \frac{1}{2}\left(\mathrm{e}^{\theta} - \mathrm{e}^{-\theta}\right) - \frac{1}{2}\left(\mathrm{e}^{\theta} + \mathrm{e}^{-\theta}\right) = -\mathrm{e}^{-\theta}$

When $\theta = 0$, $r = -1$

When $\theta = \frac{\pi}{2}$, $r = -\mathrm{e}^{-\frac{\pi}{2}}$

When $\theta = \pi$, $r = -\mathrm{e}^{-\pi}$



27) a)

When $-\frac{\pi}{2} < \theta \leq \frac{\pi}{2}$ then $0 < \cos\theta \leq 1$

So $0 < 3\cos\theta \leq 3$

Hence $0 < r \leq 3$

27) b)

When $0 < \theta \leq \frac{\pi}{2}$ then $0 < \sin\theta \leq 1$

So $0 < 2\sin\theta \leq 2$

Hence $0 < r \leq 2$

27) c)

When $0 < \theta \leq \frac{\pi}{2}$ then $0 \leq \cos\theta < 1$

So $3 \leq 3 + 2\cos\theta < 5$

Hence $3 \leq r < 5$

27) d)

When $-\pi < \theta \leq \pi$ then $-1 \leq \cos\theta \leq 1$

So $1 \leq 5 + 4\cos\theta \leq 9$

Hence $\frac{1}{3} \leq r \leq 3$

27) e)

When $0 < \theta \leq 2\pi$ then $-1 \leq \sin\theta \leq 1$

At $\theta = \frac{\pi}{2}$ then $\frac{3}{2-1} = 3$

At $\theta = \frac{3\pi}{2}$ then $\frac{3}{2+1} = 1$

The denominator is always positive

$1 \leq r \leq 3$

27) f)

When $0 \leq \theta \leq \frac{\pi}{2}$ then $0 \leq \cos\theta \leq 1$

At $\theta = 0$ then $\frac{3}{1-2} = \frac{3}{-1} = -3$

At $\theta = \frac{\pi}{2}$ then $\frac{3}{1} = 3$

$1 - 2\cos\theta = 0$ has a solution of $\theta = \frac{\pi}{3}$ for $0 \leq \theta \leq \frac{\pi}{2}$

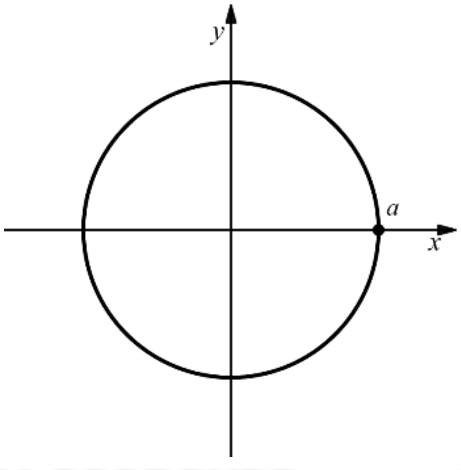
So $\frac{3}{1-2\cos\theta}$ has no solution for $\theta = \frac{\pi}{3}$

As $\theta \to \frac{\pi}{3}$, $\frac{3}{1-2\cos\theta} \to -\infty$

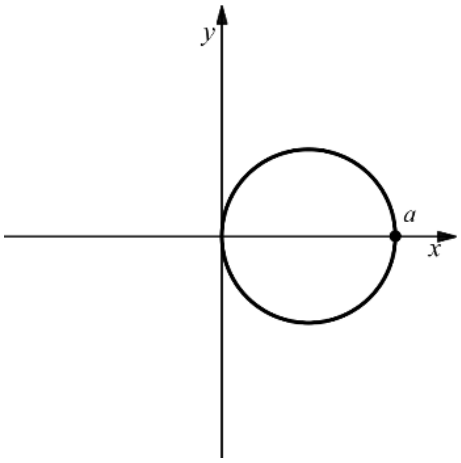
After $\frac{\pi}{3}$, θ is positive and decreasing to 3

Hence $r \leq -3$ or $r \geq 3$

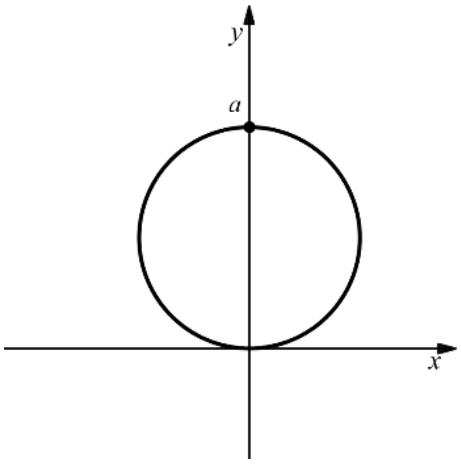
28) a)



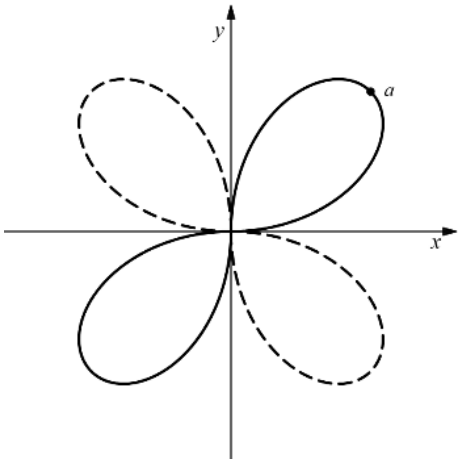
28) b)



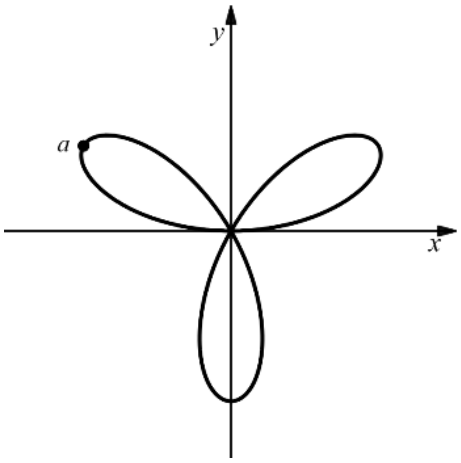
28) c)



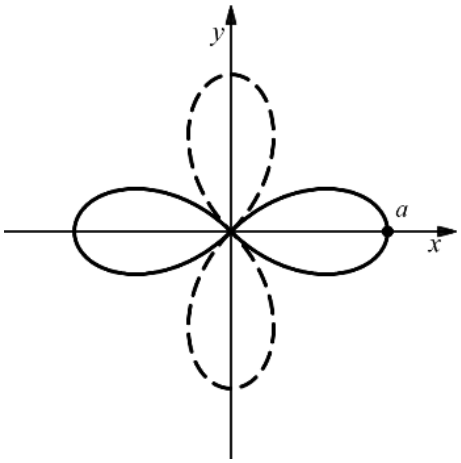
28) d)



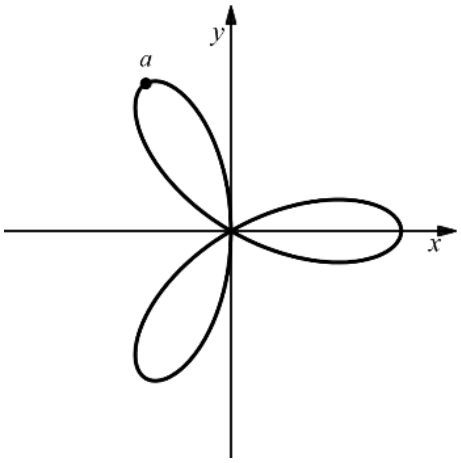
28) e)



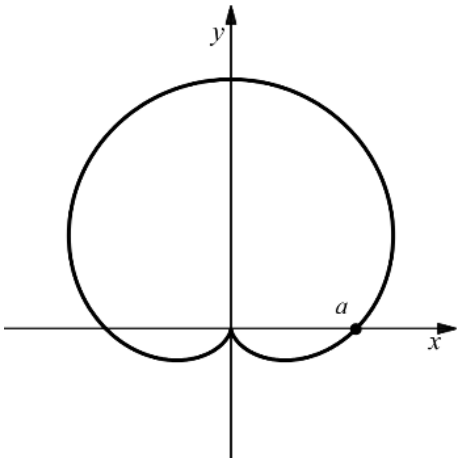
28) f)



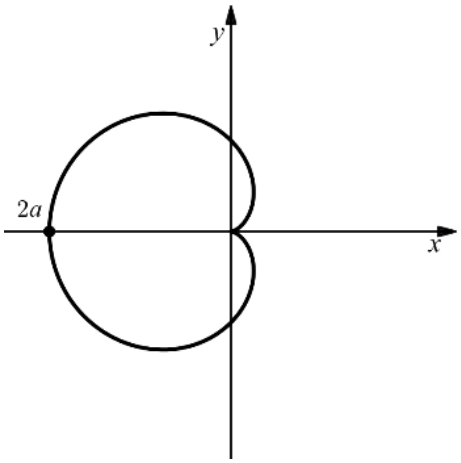
28) g)



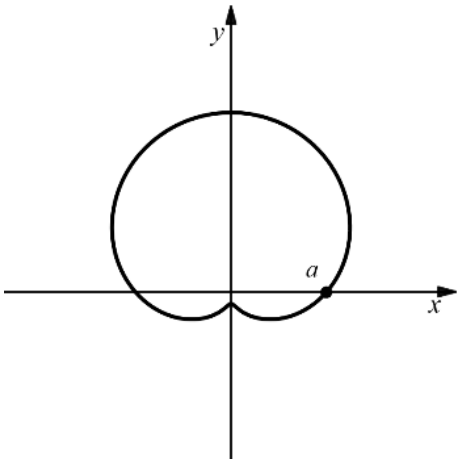
28) h)



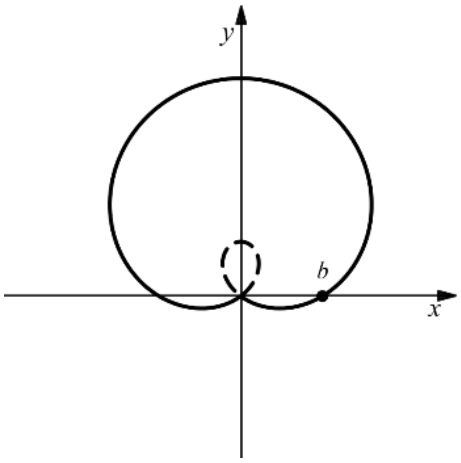
28) i)



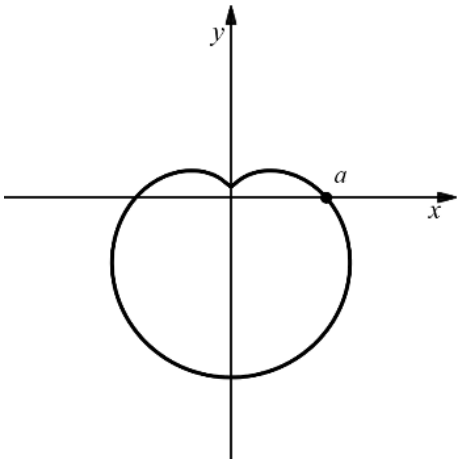
29) a)



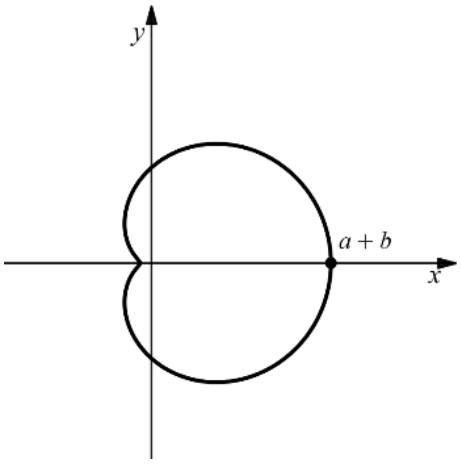
29) b)



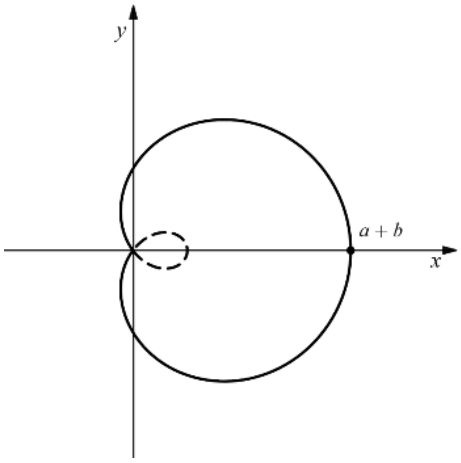
29) c)



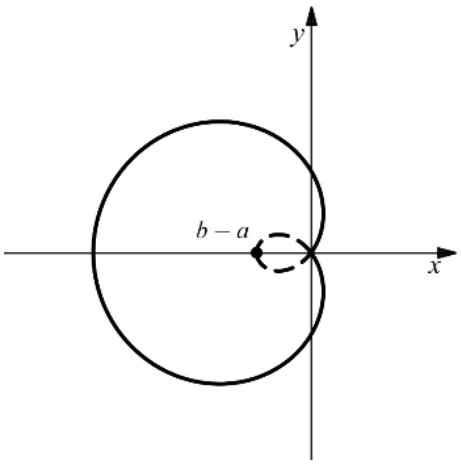
29) d)



29) e)



29) f)



30) a)

$2\theta = \theta + 2$

$\theta = 2$

So $(4, 2)$

30) b)

$3\theta = \theta + 1$

$\theta = \frac{1}{2}$

So $\left(\frac{3}{2}, \frac{1}{2}\right)$

30) c)

$\theta^2 = \theta + 2$

$\theta^2 - \theta - 2 = 0$

$\theta = -1, \theta = 2$ so $\theta = 2$

So $(4, 2)$

31) a)

$r = 2\left(1 + \cos\frac{\pi}{2}\right) = 2$

$\left(2, \frac{\pi}{2}\right)$

31) b)

$r = 2(1 + \cos 0) = 4$

$(4, 0)$

31) c)

$3 = 2(1 + \cos\theta)$

$3 = 2 + 2\cos\theta$

$\cos\theta = \frac{1}{2}$

$\theta = -\frac{\pi}{3}$ or $\theta = \frac{\pi}{3}$

$\therefore \left(3, -\frac{\pi}{3}\right)$ and $\left(3, \frac{\pi}{3}\right)$

32)

$2\cos^2\theta = 2 - 3\cos\theta$

$2\cos^2\theta + 3\cos\theta - 2 = 0$

$(2\cos\theta - 1)(\cos\theta + 2) = 0$

$\cos\theta = \frac{1}{2}$ or $\cos\theta = -2$

So $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

$\left(\frac{1}{2}, \frac{\pi}{3}\right)$ and $\left(\frac{1}{2}, \frac{5\pi}{3}\right)$

33)

$3\sin^2\theta = 2 - 3\sin\theta$

$3\sin^2\theta + 3\sin\theta - 2 = 0$

$\sin\theta = \frac{-3\pm\sqrt{33}}{6}$

$\sin\theta = \frac{-3+\sqrt{33}}{6}$ or $\sin\theta = \frac{-3-\sqrt{33}}{6}$

$(0.628, 0.475)$ and $(0.628, 2.67)$

34)

$4\sin 2\theta = 4\sin\theta$

$4(2\sin\theta\cos\theta) = 4\sin\theta$

$2\sin\theta\cos\theta = \sin\theta$

$2\sin\theta\cos\theta - \sin\theta = 0$

$\sin\theta\,(2\cos\theta - 1) = 0$

$\sin\theta = 0$ or $\cos\theta = \frac{1}{2}$

So $\theta = \pi$, $\theta = 2\pi$ or $\theta = \frac{\pi}{3}$, $\theta = \frac{5\pi}{3}$

When $\theta = \pi$, $r = 0$ but $r > 0$

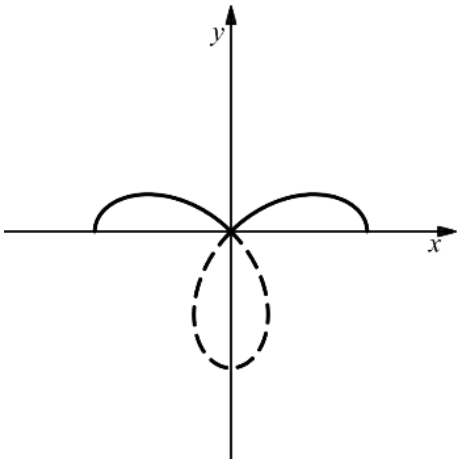
When $\theta = 2\pi$, $r = 0$ but $r > 0$

When $\theta = \frac{\pi}{3}$, $r = 2\sqrt{3}$

When $\theta = \frac{5\pi}{3}$, $r = -2\sqrt{3} \Rightarrow \theta = \frac{2\pi}{3}$, $r = 2\sqrt{3}$

$\left(2\sqrt{3}, \frac{\pi}{3}\right)$ and $\left(2\sqrt{3}, \frac{2\pi}{3}\right)$

35) a)



35) b)

$r = 3$

Expecting 2 solutions as $r = 6\cos 2\theta$ is defined for $0 \leq \theta \leq \pi$ and only 2 intersections occur when $r = 3$

The other 2 occur when $r = -3$

2 solutions from

$6\cos 2\theta = 3$

$\cos 2\theta = \frac{1}{2}$

$\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

So $\left(3, \frac{\pi}{6}\right)$ and $\left(3, \frac{5\pi}{6}\right)$

36) a)

When $\theta = \frac{\pi}{2}$ so $2a\sin^2\left(\frac{\pi}{2}\right) = 2a \times 1^2 = 2a$

When $\theta = -\frac{\pi}{2}$ so $2a\sin^2\left(-\frac{\pi}{2}\right) = 2a \times (-1)^2 = 2a$

So $P\left(2a, \frac{\pi}{2}\right)$ and $Q\left(2a, -\frac{\pi}{2}\right)$

36) b)

$2a\sin^2\theta = a(1 - \sin\theta)$

$2a\sin^2\theta + a\sin\theta - a = 0$

$2\sin^2\theta + \sin\theta - 1 = 0$

$(2\sin\theta - 1)(\sin\theta + 1) = 0$

$\sin\theta = \frac{1}{2}$ or $\sin\theta = -1$

$\theta = \frac{\pi}{6}$, $\theta = \frac{5\pi}{6}$ or $\theta = -\frac{\pi}{2}$

When $\theta = \frac{\pi}{6}$ so $2a\sin^2\left(\frac{\pi}{6}\right) = 2a \times \left(\frac{1}{2}\right)^2 = \frac{a}{2}$

When $\theta = \frac{5\pi}{6}$ so $2a\sin^2\left(\frac{5\pi}{6}\right) = 2a \times \left(\frac{1}{2}\right)^2 = \frac{a}{2}$

When $\theta = -\frac{\pi}{2}$ so $2a\sin^2\left(-\frac{\pi}{2}\right) = 2a \times (-1)^2 = 2a$

So

$\left(\frac{a}{2}, \frac{\pi}{6}\right)$, $\left(\frac{a}{2}, \frac{5\pi}{6}\right)$ and $\left(2a, -\frac{\pi}{2}\right)$